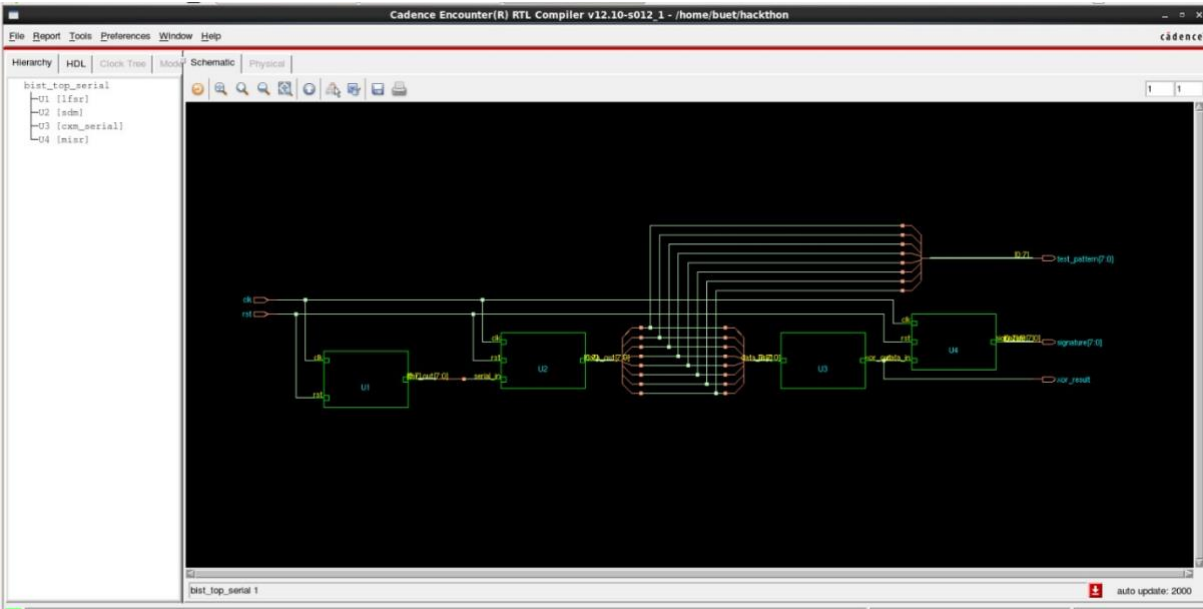
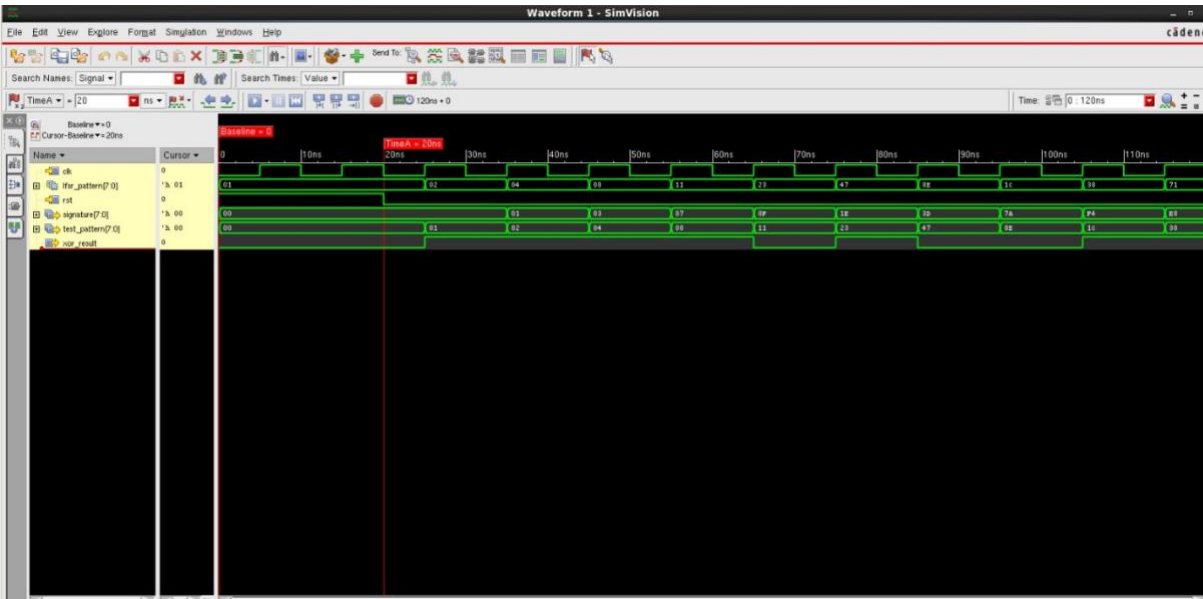


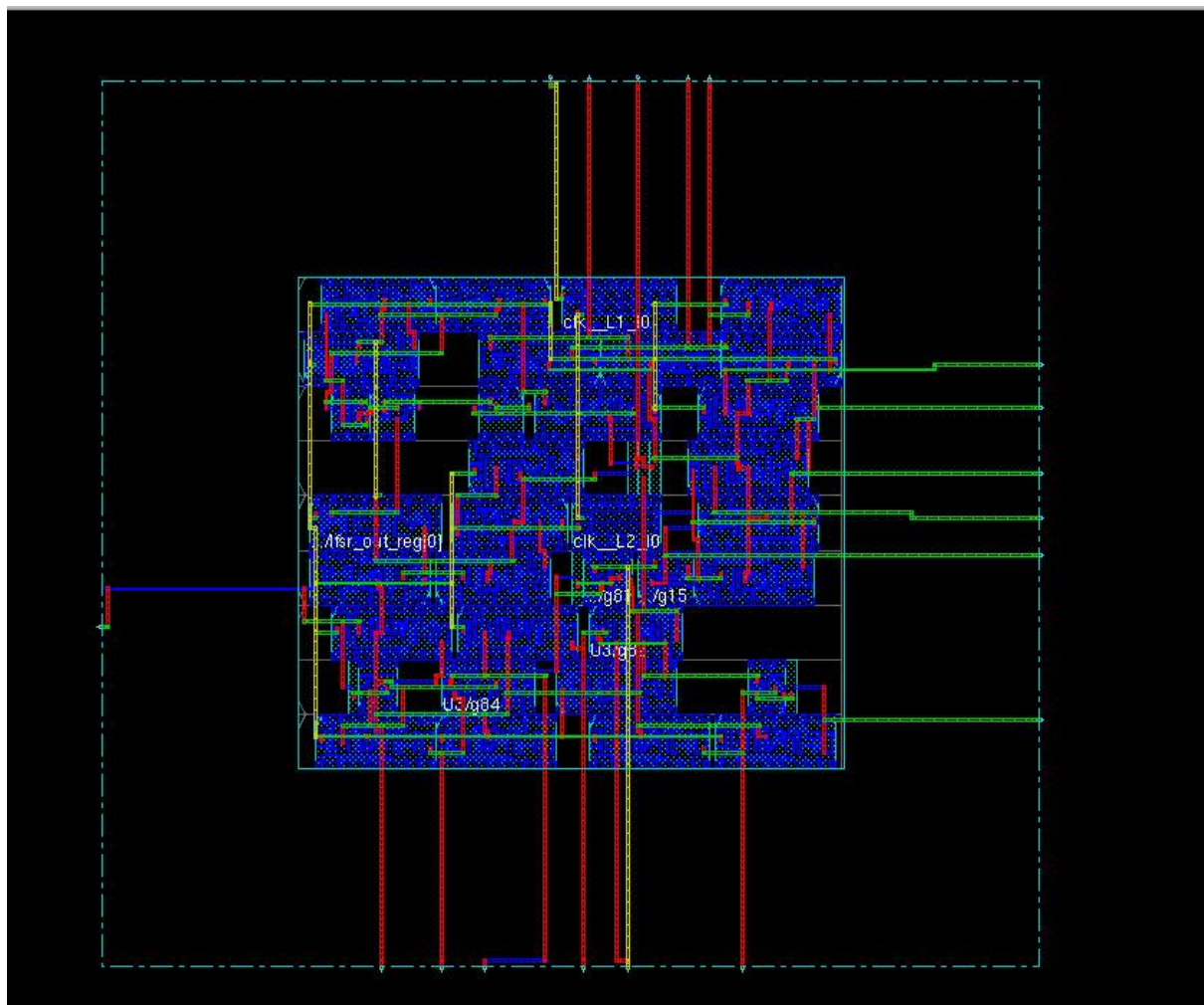
Existing



Simulation / Synthesis Report Table

Parameter	Result
Tool	Cadence RTL Compiler
Technology	SCL180 / 180nm
Top Module	bist_top_serial
Total Standard Cells	39
Total Area	1966

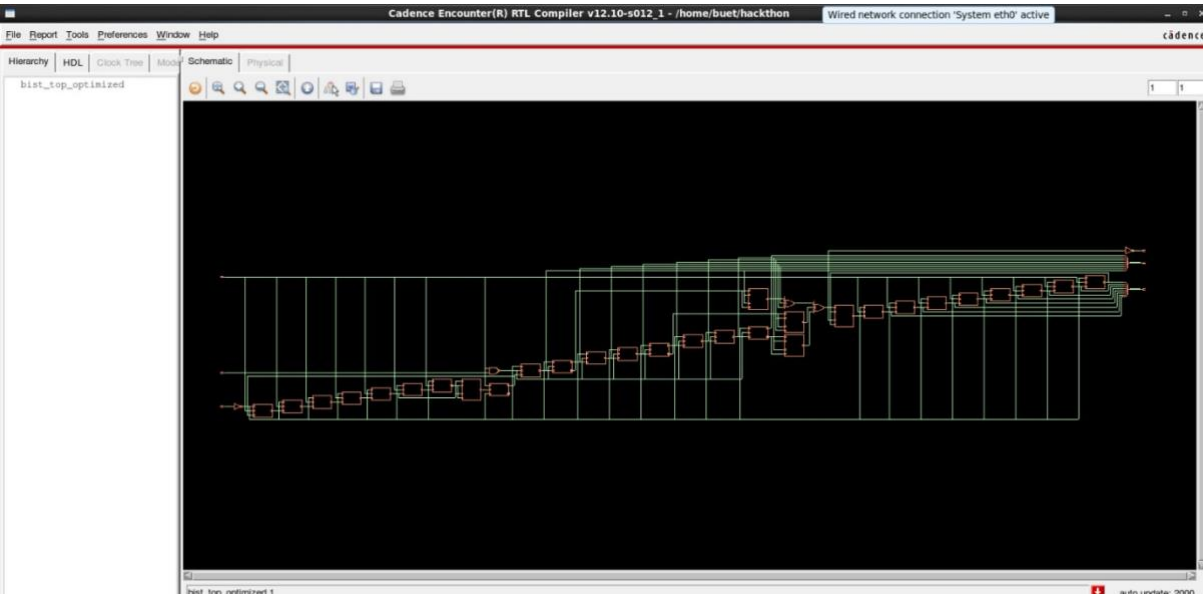
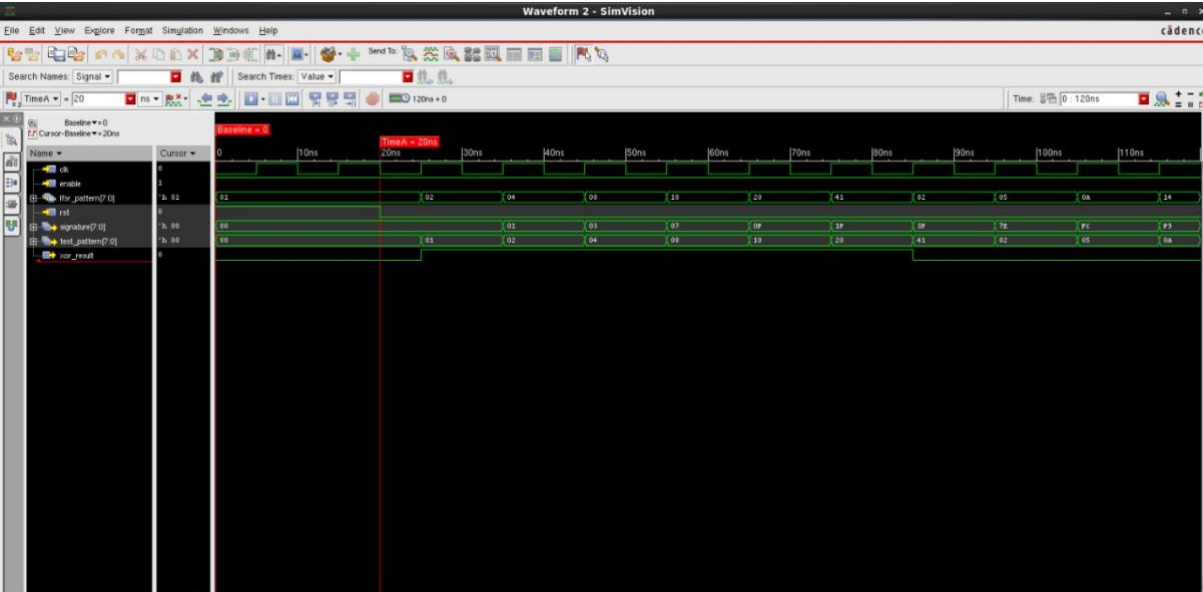
Parameter	Result
Clock Frequency	100 MHz
Clock Period	10 ns
Arrival Time	4.311 ns
Required Time	10.000 ns
Timing Slack	5.689 ns
Leakage Power	31.357 nW
Dynamic Power	353208.589 nW
Total Power	353239.946 nW
Timing Status	PASS



Physical Design (PD) Report Table

Parameter	Result
Tool	Cadence Encounter 13.10
Technology	SCL180 / 180nm
PD Stage	Post Route
Routing Status	SUCCESS
DRC Violations	0
Routing Congestion	0%
Clock Skew	3.7 ps
Total Wirelength	1463 um
Total Vias	309
Routed Nets	41
Internal Power	0.2489 mW
Switching Power	0.1275 mW
Leakage Power	0.0000338 mW
Total Power	0.3764 mW
Antenna Violations	0
Backend Status	PASS

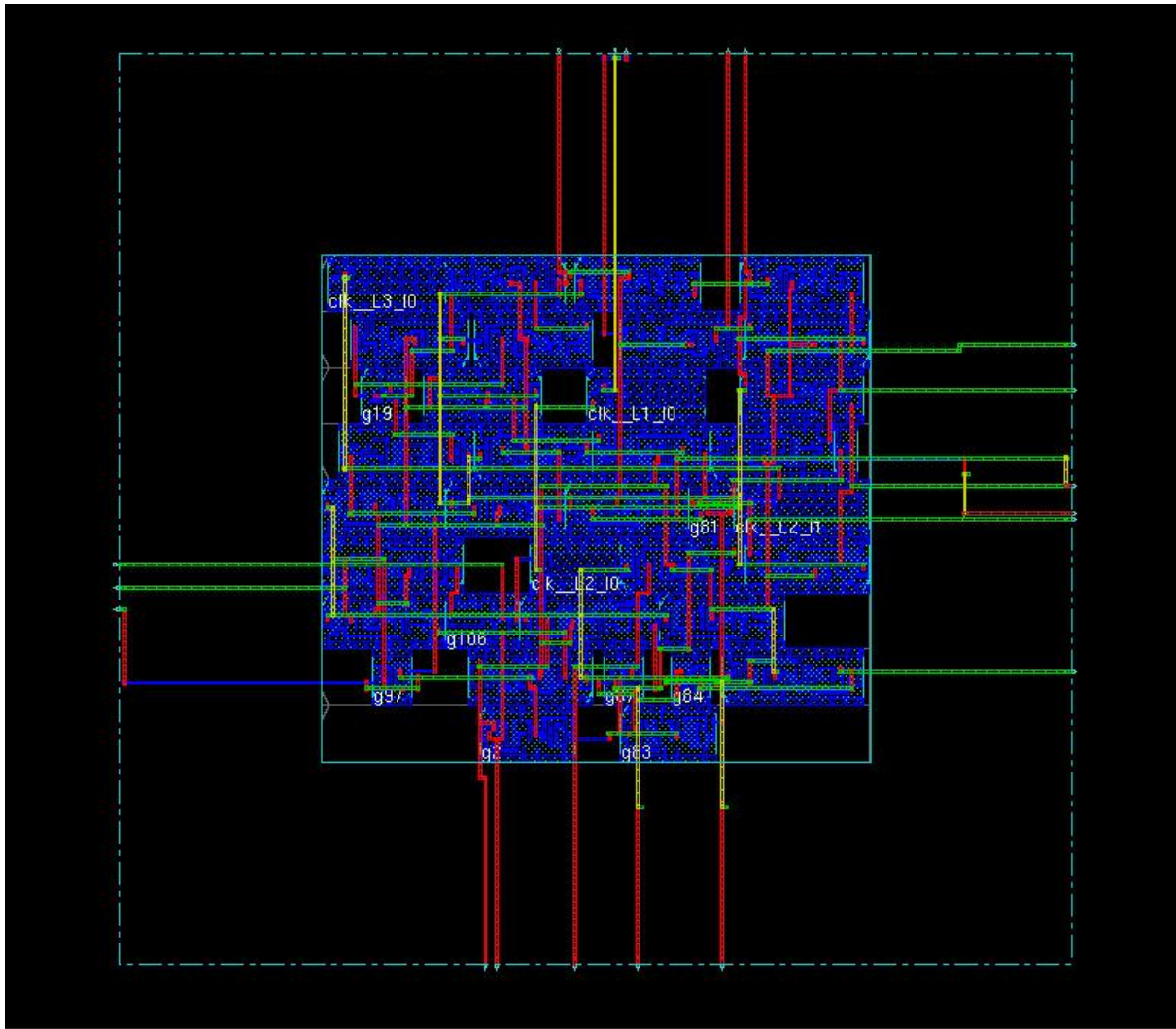
Optimized



Optimized Design — Simulation/Synthesis Results Table

Parameter	Result
Tool	Cadence RTL Compiler
Technology	SCL180 / 180nm
Top Module	bist_top_optimized
Total Standard Cells	34
Total Area	1922

Parameter	Result
Clock Frequency	200 MHz
Clock Period	5 ns
Arrival Time	3.162 ns
Required Time	5.000 ns
Timing Slack	2.138 ns
Leakage Power	30.544 nW
Dynamic Power	548265.619 nW
Total Power	548296.163 nW
Timing Status	PASS



Optimized Design — Physical Design (PD) Results Table

Parameter	Result
Tool	Cadence Encounter 13.10
Technology	SCL180 / 180nm
PD Stage	Post Route
Routing Status	SUCCESS
Total Internal Power	0.2672 mW
Total Switching Power	0.1245 mW
Total Leakage Power	0.00004416 mW
Total Power	0.3918 mW

Parameter	Result
Sequential Power	0.1851 mW
Combinational Power	0.01868 mW
Clock Network Power	0.1880 mW
Timing Slack	6.200 ns
Total Instances	38
Power Supply Voltage	1.62 V
Timing Status	PASS
Backend Status	PASS

Simulation / Synthesis Comparison Table

Parameter	Existing Design	Optimized Design	Observation
Top Module	bist_top_serial	bist_top_optimized	Optimized Architecture
Technology	SCL180 / 180nm	SCL180 / 180nm	Same Technology
Total Standard Cells	39	34	Reduced Cell Count
Total Area	1966	1922	Reduced Area
Clock Frequency	100 MHz	200 MHz	Increased Speed
Clock Period	10 ns	5 ns	Faster Clock
Arrival Time	4.311 ns	3.162 ns	Reduced Delay
Required Time	10 ns	5 ns	Higher Timing Constraint
Timing Slack	5.689 ns	2.138 ns	Timing Met Successfully
Leakage Power	31.357 nW	30.544 nW	Reduced Leakage
Dynamic Power	353208.589 nW	548265.619 nW	Increased due to Higher Frequency
Total Power	353239.946 nW	548296.163 nW	Performance-Oriented Tradeoff
Timing Status	PASS	PASS	Successful Timing Closure

Physical Design (PD) Comparison Table

Parameter	Existing Design	Optimized Design	Observation
Tool	Cadence Encounter 13.10	Cadence Encounter 13.10	Same Tool
Technology	SCL180 / 180nm	SCL180 / 180nm	Same Technology
PD Stage	Post Route	Post Route	Same Flow
Routing Status	SUCCESS	SUCCESS	Successful Routing
DRC Violations	0	0	Clean Layout
Routing Congestion	0%	0%	No Congestion
Timing Slack (PD)	6.311 ns	6.200 ns	Similar Timing Margin
Total Internal Power	0.2489 mW	0.2672 mW	Slight Increase
Total Switching Power	0.1275 mW	0.1245 mW	Reduced Switching Power
Total Leakage Power	0.0000338 mW	0.00004416 mW	Slight Increase
Total Power	0.3764 mW	0.3918 mW	Slight Increase
Sequential Power	0.2651 mW	0.1851 mW	Reduced Sequential Power
Combinational Power	0.02986 mW	0.01868 mW	Reduced Logic Power
Clock Network Power	0.08142 mW	0.1880 mW	Increased due to 200MHz Clock
Backend Status	PASS	PASS	Full ASIC Flow Completed